

Lace and Tie a Shoe SOP (1x Episode: 1 Shoe)

One episode laces and ties **one** shoe. The shoe is carried from the input pile to the working area, squared up sole down with the toe away from the front edge, the lace is freed and evened, the bottom eyelet pair is threaded, the lace is crossed up one pair at a time to the top pair, every cross is pulled snug from the bottom up, a bow is tied and confirmed, both ends are tucked away, and the finished shoe is set down in the output zone.

The order never changes: straighten first, thread from the bottom pair up, tension from the bottom cross up, tie, tuck, then place. No eyelet is threaded before the shoe is squared and the lace is free of every eyelet, and no pair is skipped or threaded twice.

Every pair above the bottom carries one **cross**, with the same end sitting on top of every X. The bottom pair goes in from above and every eyelet after it is entered from underneath.

The left gripper carries the shoe in, holds it steady, centers the tongue, and works the far eyelet row. The right gripper works the near eyelet row, leads the search for a lace end, leads the knot and the bow, and carries the finished shoe to the output zone.

Setup

Complete both checklists before starting an episode.

Cell configuration

- **Environment camera:** 900 mm.
- **cell_type:** bimanual

Hardware checklist

1. Cameras are on and recording.
2. The environment camera shows the whole table: the shoe pile at the back left, the working area at the center, and the output zone at the back right.
3. The working area is visible from above, so both eyelet rows, the tongue, and every cross can be seen.
4. Both arms are at home with grippers open.
5. The table is clean, dry, and bare apart from the shoes.
6. The left arm reaches the shoe pile and the working area. The right arm reaches the working area and the output zone.

Materials checklist

1. One or more **shoes** sit at the back left, sole down. Each one is unlaced, with its **lace** lying loose across the **tongue**, both **lace ends** free, and every **eyelet** empty.
2. Every shoe has both **aglets** sound, no torn eyelet, and a tongue that lies flat when it is pushed into place.
3. Each lace is long enough to cross every eyelet pair and still tie a bow with tails to spare.
4. The working area at the center is clear.
5. The output zone at the back right is clear, or holds finished shoes from earlier episodes in one row, all toes pointing the same way.

Workspace layout

- **Shoe pile:** back left, the unlaced shoes lying sole down.
- **Working area:** center, where the straightening, threading, tensioning, tying, and tucking happen.
- **Output zone:** back right, the laced shoes in one row.

Everything the left arm picks up is staged on the left. The right arm works the center and the right. Neither arm reaches across to the other side of the table.

Arm assignments

- **Left gripper:** carries the shoe in from the shoe pile, steadies it during the straighten, centers the tongue, holds the far **aglet**, threads and crosses the **far eyelet row**, and forms the first loop of the bow.
- **Right gripper:** squares the shoe in the working area, leads the search for a lace end, holds the near **aglet**, threads and crosses the **near eyelet row**, forms and passes the second loop, tucks the near end, and carries the finished shoe to the output zone.

Vocabulary

- **Shoe:** one lace-up shoe, laced and tied whole in a single episode.
- **Toe:** the front end of the shoe, farthest from the ankle opening. It points away from the front edge of the table for the whole episode.
- **Heel:** the back end of the shoe, behind the ankle opening. Every carry grasps the shoe here.
- **Collar:** the ankle opening at the heel end.
- **Tongue:** the flap between the two eyelet rows, under the lace.
- **Eyelet:** one of the holes the lace passes through.
- **Eyelet row:** the line of eyelets down one side of the opening. The **near row** is the one toward the front edge of the table and the **far row** is the one away from it.

- **Eyelet pair:** the two facing eyelets at the same height, one in each row.
- **Bottom eyelet pair:** the pair closest to the toe. Threading starts here. **Top eyelet pair:** the pair closest to the collar. Threading ends here.
- **Lace:** the single cord threaded through the eyelets.
- **Aglet:** the stiffened tip at either end of the lace. Every grasp of a lace end closes on the aglet, never on a bight of lace in the middle.
- **Lace end:** the free length of lace running from an aglet back to the last eyelet it passed through.
- **Even ends:** the two lace ends look the same length when both aglets are held up side by side.
- **Bridge:** the flat run of lace between the two bottom eyelets, lying across the tongue.
- **Cross:** one X of lace, made where the two ends pass each other on their way to the opposite row. One cross sits at every pair above the bottom pair.
- **Same end on top:** at every cross, the same lace end lies over the other one, so the X pattern looks the same all the way up.
- **Snug cross:** a cross pulled flat against its eyelets with no slack showing, but not so tight that the tongue bunches.
- **Even opening:** the two eyelet rows look parallel and the gap between them looks the same from the bottom pair to the top pair.
- **Starting knot:** the single knot tied just above the top eyelet pair, seated down on the top cross before the loops are formed.
- **Loop:** one of the two folded bights of lace that make the bow. Both loops look the same size, each about as wide as the shoe opening, and both sit across the opening rather than along it.
- **Bow:** the finished two-loop knot. It holds when both loops are pulled apart.
- **Tucked end:** a lace end pushed under the nearest cross or inside the collar, so nothing hangs below the top of the sole and nothing drapes onto the table.
- **Lift:** coming straight down onto the grasp point, closing, and raising straight up until there is daylight under the thing, before anything moves sideways.
- **Release point:** over the working area or over the output zone. A gripper opens nowhere else.

Steps

Step 1 brings the shoe in and Step 2 straightens it. Steps 3 to 5 build and tighten the lacing. Steps 6 and 7 tie and tidy it. Step 8 puts the shoe in the output zone and Step 9 ends the episode.

Step 1: Bring one shoe to the working area

Goal: one shoe sits sole down in the working area, ready to be straightened.

- The **left gripper** closes on one **shoe** in the shoe pile at the **heel**, lifts it clear of the table, carries it to the working area, sets it down sole down at the center, and releases.

Take one shoe only, by the heel. Never drag or slide a shoe across the table, and never lift a shoe by its lace or its tongue.

Expected state: one shoe stands sole down in the middle of the working area with both eyelet rows in view from above.

Step 2: Straighten the shoe and free the lace

Goal: the shoe sits flat and sole down with the toe pointing away from the front edge, the tongue flat and centered, every eyelet empty, and both lace ends free and even.

2.1 Square the shoe

- The **right gripper** closes on the **heel** and turns the shoe until the **toe** points away from the front edge and the two eyelet rows run front to back, then releases.
- If the toe points toward the front edge, turn the shoe a half turn so the toe faces away and the collar faces the front edge.
- If the shoe lies on its side or sole up, the **left gripper** takes the **collar** and the **right gripper** takes the **toe**, both lift the shoe clear of the table, roll it upright in one smooth motion, set it down sole down, and release. If it topples, 2.1 runs again.

Expected state: the shoe stands sole down, toe away from the front edge, with the **near row** and the **far row** both in view from above.

2.2 Find and hold the two lace ends

- The **right gripper** closes on one **aglet**.
- If no aglet is in view, the **right gripper** closes on the highest point of the lace, turns the lace anti-clockwise until an aglet comes into view, releases, and closes on that aglet.
- The **left gripper** closes on the second **aglet** the same way.

Always search anti-clockwise, and always close on an aglet. Never treat a bight of lace in the middle as an end.

2.3 Clear the eyelets and free the lace

- Both grippers draw their ends up and away from the shoe until every **eyelet** is empty and no lace is left threaded.
- Look over the tongue, both eyelet rows, and the collar. Lift any lace trapped under the tongue or hooked on an eyelet clear, then lay it back outside that row.
- Lay the lace across the tongue with one end resting outside each eyelet row.

Expected state: every eyelet is empty from above and both lace ends run free with nothing snagged.

2.4 Center the tongue

- The **left gripper** closes on the top edge of the **tongue** and draws it up toward the **collar** until it lies flat and centered between the two eyelet rows, then releases.

2.5 Even the lace ends

- The **right gripper** holds the near **aglet** and the **left gripper** holds the far aglet.
- Bring both aglets up to the same height and pull outward in opposite directions until the twists and the slack come out.
- Draw the longer end back until the two ends are **even ends**.

Check: the shoe stands sole down with the toe away from the front edge, the tongue lies flat and centered, every eyelet is empty, and the two lace ends are even with no twists.

Step 3: Thread the bottom eyelet pair

Goal: the lace is seated in the bottom eyelet pair with a flat bridge across the tongue and the two ends still even.

- The **right gripper** holds the near **aglet** and the **left gripper** holds the far aglet.
- The **right gripper** passes its end **down** through the near **bottom eyelet** from above until the aglet comes out underneath.
- The **left gripper** passes its end **down** through the far bottom eyelet from above the same way.
- Both grippers draw their ends until the **bridge** between the two bottom eyelets lies flat across the tongue, then draw the longer end back until the ends are even again.

The bottom pair is the only pair entered from above. Never start at the top pair or at a middle pair, and never work downward toward the toe.

Check: the bottom pair carries one flat bridge across the tongue, both ends come out underneath the shoe, and the two ends are even.

Step 4: Cross the lace up to the top pair

Goal: every pair above the bottom pair carries one cross, with the same end on top at every X, and both ends leaving the top pair, one per row.

- The **right gripper** takes the end coming out from under the **near row**, carries it diagonally across the tongue to the next eyelet up in the **far row**, threads it **up** through that eyelet from underneath, and draws it clear.
- The **left gripper** takes the end coming out from under the far row, carries it diagonally to the next eyelet up in the **near row**, threads it **up** through that eyelet from underneath, and

draws it clear.

- Keep the two ends crossing in the same order at every pair, so the **same end on top** at every X.
- Repeat one pair at a time, working from the bottom pair up. Do not skip a pair and do not thread a pair twice.
- Stop when both ends leave the **top eyelet pair**, one end per row.

Every eyelet above the bottom pair is entered from underneath. Cross the tongue in the air, without dragging the aglet across the tongue, and finish one pair before starting the next.

Check: every pair between the bottom pair and the top pair carries exactly one cross, no pair is empty and none carries two, the same end sits on top of every X, and one end leaves each row at the top pair.

Step 5: Tension the crosses from the bottom up

Goal: every cross snug, the opening even from the bottom pair to the top pair, and the two ends even.

- The **right gripper** holds one end and the **left gripper** holds the other, at the bottom cross.
- Pull the two grippers apart until that cross is a **snug cross**.
- Work up one cross at a time in the same way, so the tension comes on evenly along the whole opening.
- Adjust until the opening is an **even opening**.
- Draw the longer end back until the two ends are even.
- If a cross stays slack, an end pulls back out of an eyelet, or the tongue bunches, return to Step 4 for that pair and rethread it.

Tension bottom to top, one cross at a time. Never pull only at the top pair and call the opening done.

Check: every cross lies flat against its eyelets with no slack, the two rows look parallel with an even gap from bottom to top, the tongue lies flat and unbunched, and the two ends are even.

Step 6: Tie a bow that holds

Goal: the shoe carries a bow with two loops the same size and a knot that holds under a pull.

6.1 Tie the starting knot

- The **right gripper** holds the near end and the **left gripper** holds the far end, both just above the **top eyelet pair**.
- Cross the near end over the far end, pass it under and through, then pull both grippers apart until the **starting knot** seats down on the top cross.

6.2 Form the first loop

- The **left gripper** folds the far end back on itself into a **loop** and pinches it at its base, directly above the starting knot.

6.3 Wrap and pass the second loop

- The **right gripper** carries the near end around the base of the first loop.
- Push the fold of the near end through the gap under that wrap to form the second **loop**.

6.4 Pull the bow tight and confirm it

- Each gripper takes one loop and both pull apart until the knot seats and both loops hold their shape.
- Adjust until the two loops look the same size and both sit across the opening rather than along it.
- Pull both loops apart once more. If the knot slips, loosens, or comes undone, return to 6.1 and tie it again.
- Release both grippers.

Check: the starting knot sits down on the top cross, the bow holds under the confirming pull, both loops look the same size, and both lie across the opening.

Step 7: Tuck the two lace ends

Goal: no lace end hangs loose. Each end sits under a cross or inside the collar, clear of the table.

- The **right gripper** closes on the near end below the bow and pushes it under the nearest **cross** or inside the **collar** until it is a **tucked end**, then releases.
- The **left gripper** does the same with the far end.
- If an end works loose, tuck it again.

Check: no lace end and no loop hangs below the top of the sole, nothing drapes onto the table, and the bow is still tight with both loops across the opening.

Step 8: Move the laced shoe to the output zone

Goal: the laced shoe stands sole down in the output zone at the back right.

- The **right gripper** closes on the laced shoe at the **heel**, lifts it clear of the table, and carries it back and to the right into the **output zone**.
- If a shoe is already there, set the new shoe beside it with a clear gap, toe pointing the same way, not touching.
- Lower it until it sits sole down, then release.

Carry the shoe by the heel only. Never close on the lace, the bow, or a loop, never set a shoe on top of another shoe, and never drag the finished shoe across the table.

Check: the laced shoe stands sole down in the output zone in one row with any earlier shoes, all toes pointing the same way, the bow is still tight, and both ends are still tucked.

Step 9: Confirm the lacing and end the episode

- Confirm the shoe carries one cross at every pair from the bottom pair to the top pair, the same end on top at every X, an even opening, a bow with two loops the same size that held under the pull, both ends tucked with nothing hanging below the sole, the shoe in the output zone sole down, and the working area clear.
- Return both arms home, clear of the working area and the output zone, then stop recording.

After the episode: reset the workspace

This reset is not recorded.

Between episodes of a batch

1. Leave the laced shoe in the output zone and the remaining shoes at the back left.
2. Clear the working area and confirm no lace end or aglet is left on the table.
3. Run the Setup checklists again before the next episode.

At the end of a batch

1. Carry the laced shoes from the output zone back to the shoe pile at the back left.
2. Untie each bow and unthread the lace from every eyelet.
3. Lay each lace loose across the tongue with both ends free and every eyelet empty, so the input pile matches the setup state.
4. Set every shoe sole down at the back left.
5. Wipe the table and confirm the surface is clean, dry, and bare apart from the shoes.
6. Replace a shoe with a torn eyelet or a tongue that will not lie flat, and a lace with a frayed or split aglet or one too short to cross every pair and still tie a bow.

SOP violations

These are actions that break the SOP and are reviewed side by side in the review tool.

How to record a violation in review

For each violation, record the **start timestamp**, **violation name**, and **SOP rule broken**. The visible cue is what the reviewer sees. The coaching note is for retraining and is not an annotation label.

Episode handling

Tag every violation with its timestamp and name. Keep the episode with the violation tag. Do not delete it just because a rule was broken.

Violations

Violation: Wrong pickup

- **Visible cue:** the shoe is taken from the working area, the right side, or the output zone instead of the shoe pile, more than one shoe is lifted in one grasp, the shoe is lifted by its lace or its tongue rather than the heel, it is dragged or slid across the table instead of lifted and carried, or it is set down outside the working area.
- **SOP rule broken:** Step 1, the left gripper lifts one shoe out of the shoe pile by the heel and sets it down sole down in the working area.
- **Coaching note:** one shoe per episode, taken by the heel, lifted clear of the table and carried.

Violation: Shoe not straightened before threading

- **Visible cue:** threading starts while the shoe lies on its side or sole up, while the toe points toward the front edge, while the tongue is folded under, pushed to one side, or bunched, or while lace is still threaded through an eyelet or hooked under the tongue from the last episode.
- **SOP rule broken:** Steps 2.1, 2.3, and 2.4, square the shoe sole down with the toe away from the front edge, clear every eyelet, and center the tongue before any threading.
- **Coaching note:** square, clear, and center first. A skewed tongue or leftover lace pulls the whole pattern off.

Violation: Lace ends not prepared

- **Visible cue:** a gripper closes on empty air or on a bight of lace in the middle and that point is worked as an end, the search for a hidden aglet turns clockwise or back and forth instead of anti-clockwise, or threading starts with the two ends visibly different lengths.
- **SOP rule broken:** Steps 2.2 and 2.5, close on an aglet, search anti-clockwise when no aglet is in view, and even the two ends before the bottom pair is threaded.
- **Coaching note:** confirm the aglet is in view before you close. Even the ends before the first thread, not after the top pair.

Violation: Wrong threading start

- **Visible cue:** the first pair threaded is not the bottom pair, the lacing starts at the top pair or a middle pair, or it works downward toward the toe.
- **SOP rule broken:** Step 3, seat the lace in the bottom eyelet pair first and work up from there.

- **Coaching note:** bottom pair first, every time, then one pair up at a time.

Violation: Threaded from the wrong side of an eyelet

- **Visible cue:** the bottom pair is entered from underneath instead of from above, or an eyelet above the bottom pair is entered downward from above instead of up from underneath.
- **SOP rule broken:** Step 3, the bottom pair goes in from above, and Step 4, every eyelet above it is entered up from underneath.
- **Coaching note:** bottom pair down from the top, every cross after that up from below.

Violation: Pair skipped or threaded twice

- **Visible cue:** at the end of Step 4 a pair between the bottom and the top carries no lace, or a pair carries two passes, so the pattern is uneven up the opening.
- **SOP rule broken:** Step 4, one cross per pair, working bottom to top, with no pair skipped and none threaded twice.
- **Coaching note:** finish one pair and look at it before you move up.

Violation: Not a cross pattern, or crosses inconsistent

- **Visible cue:** the lace runs straight across a pair or up the same row instead of each end going to the opposite row, so no X forms, or the end sitting on top of the X changes from pair to pair.
- **SOP rule broken:** Step 4, each end crosses to the opposite row at the next eyelet up, with the same end on top at every X.
- **Coaching note:** every end goes to the other row, and the same end lies on top all the way up.

Violation: Uneven tension

- **Visible cue:** at the end of Step 5 some crosses are slack while others are pulled tight, the two eyelet rows are not parallel, the gap between them is visibly wider at one end of the opening, the tongue bunches, or only the top pair was pulled.
- **SOP rule broken:** Step 5, pull each cross snug in turn from the bottom cross up until the opening is even.
- **Coaching note:** work cross by cross from the bottom. Pulling only at the top leaves slack below.

Violation: Bow does not hold

- **Visible cue:** the starting knot never seats down on the top cross, or the bow slips, loosens, or comes undone on the confirming pull and the episode carries on.
- **SOP rule broken:** Steps 6.1 and 6.4, seat the starting knot on the top cross, then confirm the bow holds under a pull and retie it if it opens.

- **Coaching note:** seat the knot firmly before the loops, and tie it again if the confirming pull opens it.

Violation: Loops uneven or wrongly set

- **Visible cue:** at the end of Step 6 the two loops are visibly different sizes, one loop is collapsed, or the loops lie along the opening instead of across it.
- **SOP rule broken:** Steps 6.2 to 6.4, two loops the same size, each about as wide as the shoe opening, both sitting across the opening.
- **Coaching note:** match the loops and set them across the opening before releasing.

Violation: Ends not tucked

- **Visible cue:** at the end of Step 7 an aglet or a loop hangs below the top of the sole, an end drapes onto the table, or an end works loose and is left that way.
- **SOP rule broken:** Step 7, each end is pushed under the nearest cross or inside the collar, clear of the table.
- **Coaching note:** tuck both ends and check nothing hangs past the sole line before the shoe moves.

Violation: Laced shoe placed wrong

- **Visible cue:** at Step 8 the shoe is moved with the left gripper, is not grasped by the heel, is gripped by the lace or the bow, is left in the working area, is dragged to the output zone, lands well off the zone, is set down on its side, is set on top of another shoe, or is set touching the shoe beside it or pointing the opposite way.
- **SOP rule broken:** Step 8, the right gripper lifts the shoe by the heel and carries it into the output zone, sole down, beside any shoe already there with a clear gap, toe pointing the same way and never stacked.
- **Coaching note:** finished shoes go in a row facing the same way, sole down. Do not pile them.

Violation: Arms swapped roles

- **Visible cue:** the right gripper carries a shoe in from the shoe pile or centers the tongue, or the left gripper squares the shoe, threads the near row, forms the second loop, or carries the finished shoe to the output zone.
- **SOP rule broken:** Steps 1 to 8, the left gripper brings the shoe in, centers the tongue, and works the far row; the right gripper squares the shoe, works the near row, leads the bow, and carries the shoe out.
- **Coaching note:** each gripper owns its own row. Nothing crosses to the other side of the table.

Violation: Dropped, collided, or knocked over

- **Visible cue:** the shoe falls to the table or the floor short of its spot, the shoe topples in the working area, a shoe already in the output zone is struck and shifts or tips, or the two arms strike each other.
- **SOP rule broken:** Steps 1 to 8, keep the shoe on a clear travel path and release only after a settled placement.
- **Coaching note:** check the path and the landing spot before you move.

Violation: Wrong episode ending

- **Visible cue:** the episode ends with an eyelet pair unlaced, a slack cross, no bow, a bow left untested, an end untucked, the shoe still in the working area, the shoe on its side in the output zone, or an arm away from home.
- **SOP rule broken:** Step 9, confirm the lacing, the bow, the tucked ends, and the placed shoe; return both arms home clear of the scene; then stop recording.
- **Coaching note:** confirm first. Homing is the last thing the arms do.

Non-violation failures

These failures are not caused by how the task was run. Log them as system issues, discard the episode, and never use them for coaching.

- **Recording stopped or paused during the episode** (recording system).
- **Camera dropped frames or lost its feed** (capture system).
- **Hardware fault on an arm:** gripper failure, drift, controller caused collision, or motor error.
- **Defective shoe:** a torn eyelet, an eyelet that will not pass the lace, or a tongue that will not lie flat under a correct pull. Replace it before the next episode.
- **Defective lace:** a frayed or split aglet, or a lace too short to cross every pair and still tie a bow. Replace it before the next episode.

Annotation subtasks (from SOP)

1. Move one shoe from the shoe pile to the working area
2. Square the shoe sole down with the toe away from the front edge
3. Find both lace ends, clear every eyelet, and center the tongue
4. Even the two lace ends
5. Thread the bottom eyelet pair
6. Cross the lace up to the top eyelet pair
7. Tension the crosses from the bottom up
8. Tie the starting knot and form the two loops
9. Tuck the two lace ends
10. Move the laced shoe to the output zone

11. Confirm the lacing and end the episode